

Catalyst Switch Lab: Small Office Network Build Guide

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Overview

You'll build a two-switch small-office network on your real Catalyst switches, break it on purpose, and publish the configs, verification output and test results on GitHub. It proves two things at once: you can configure real Cisco hardware over a console cable, and you can explain what you built in writing.

The lab covers what entry-level networking interviews ask about: VLANs, 802.1Q trunking, LACP EtherChannel, Rapid PVST+, SSH-only management, DHCP, syslog and config backups. If one switch is Layer 3, it adds inter-VLAN routing.

Item	Notes
2 Catalyst switches	Already have them
Console cable	USB-to-RJ45 console cable, or a rollover cable plus a USB-to-serial adapter
Laptop	PuTTY (Windows) or screen (macOS/Linux) at 9600 baud
4 to 6 Ethernet cables	Two for the EtherChannel, the rest for test devices
Your Ubuntu server	Becomes the lab's syslog and TFTP backup server

Two rules before you start. Keep the lab off your home network: the lab runs its own DHCP, and a second DHCP server on your home LAN will break things for everyone in the house. And write notes as you go, including mistakes, because the troubleshooting log is the most convincing part of the final write-up.

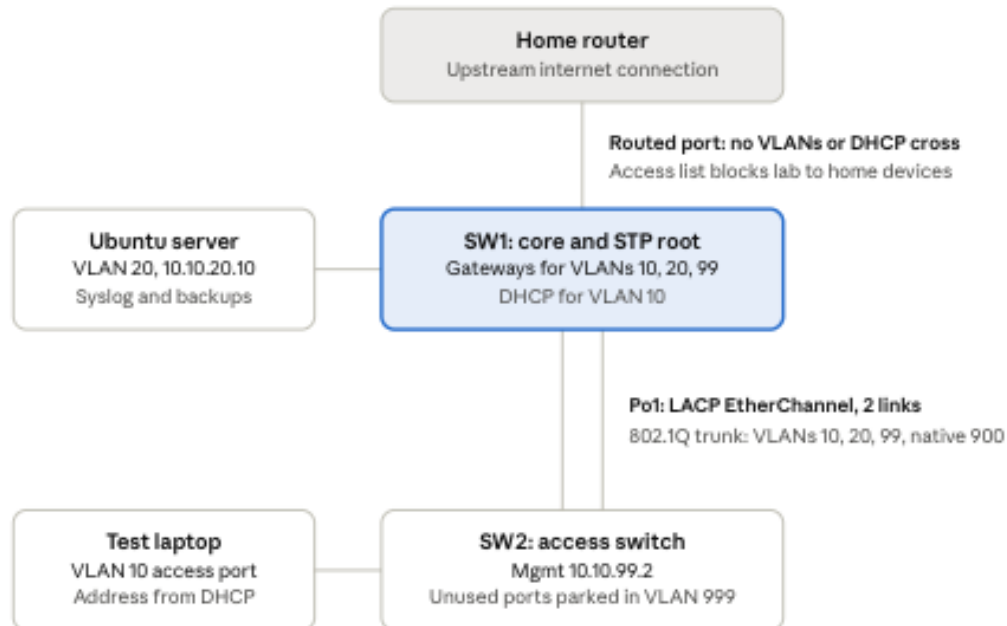
First check: run `show version` on each switch. The model tells you whether it can route (a 2960 is Layer 2 only; a 3560, 3650, 3750 or 9300 can route), and the software line tells you whether it runs IOS or IOS XE. Phases 4 and 5 below depend on the answer.

Topology and addressing

SW1 acts as the core switch, STP root and (if it can route) the gateway for every VLAN; SW2 is the access switch your test laptop plugs into.



Lab traffic reaches the internet through one routed port on SW1



lab topology · 2 switches, 2 test devices

The two uplinks between the switches form one logical link, so pulling either cable should not drop traffic. If neither switch can route, move the server into VLAN 99 so everything shares one subnet.

VLAN	Name	Subnet	Gateway	Used for
10	USERS	10.10.10.0/24	10.10.10.1 (SW1)	Test laptop; DHCP leases from .21
20	SERVERS	10.10.20.0/24	10.10.20.1 (SW1)	Ubuntu server, static 10.10.20.10
99	MGMT	10.10.99.0/24	10.10.99.1 (SW1)	Switch management: SW1 .1, SW2 .2
900	NATIVE	none	none	Trunk native VLAN, carries no hosts
999	PARKING	none	none	Unused ports, shut down

Build phases

Each phase gives you the goal, how to tell it's working, and what to research. Working out the commands yourself is the point. Keep a running notes file of what you tried, what failed and what fixed it, because that becomes your troubleshooting log. Save the verification output after each phase for the GitHub repo.

Phase 0: Reset and inventory

Goal: record each switch's model, software version, Layer 2 or Layer 3 capability and port layout, then wipe the old configuration, including the saved VLAN database.

- Done when: both switches boot with no configuration and your inventory notes are saved.
- Research: how to fully erase a Catalyst switch, and the password recovery procedure for your model if an old password locks you out.

Phase 1: Secure baseline

Goal: each switch has a hostname, an encrypted privileged-mode password, a local admin account, a login on the console, a login banner, and remote access by SSH version 2 only, with Telnet disabled.

- Done when: the console asks for a username, and (after Phase 4) SSH works while Telnet is refused.
- Research: what a switch needs configured before it can generate SSH keys, and the difference between an enable password and an enable secret.

Phase 2: VLANs, trunk and EtherChannel

Goal: create the VLANs from the addressing table on both switches. Bundle the two uplinks into one LACP port-channel and make it an 802.1Q trunk that allows only the lab VLANs, uses VLAN 900 as native and doesn't negotiate trunking. Set test-device ports as access ports with PortFast and BPDU Guard, and shut down every unused port in VLAN 999.

- Done when: the port-channel is up with both links bundled, the trunk carries only the allowed VLANs, and every port is either in use or shut down in VLAN 999.
- Research: LACP active versus passive mode, where trunk settings belong once ports are in a channel, and whether your model needs the trunk encapsulation set manually.

Phase 3: Spanning tree

Goal: run Rapid PVST+ on both switches, with SW1 deliberately the root for every lab VLAN and SW2 as the backup root.

- Done when: SW1 reports itself as root for each VLAN, and you can explain which ports are blocking and why.
- Research: how bridge priority works and how the root is elected when you don't set it.

Phase 4: Management and routing

Goal: give SW2 a management address in VLAN 99 and a way to reach other subnets. If SW1 is Layer 3, enable routing and give it a gateway address in each VLAN from the table. If neither switch routes, give SW1 only its management address and move the server into VLAN 99.

- Done when: you can SSH to both switches from your laptop and, on a Layer 3 SW1, a device in VLAN 10 can reach the server in VLAN 20.
- Research: switched virtual interfaces (SVIs), how a Layer 2 switch reaches other subnets compared with a Layer 3 switch, and SDM templates if routing won't enable.

Phase 5: DHCP for the user VLAN

Goal: SW1 hands out VLAN 10 addresses automatically, reserving .1 through .20 and giving clients the right default gateway.

- Done when: the test laptop gets a lease on its own and the switch shows the binding.
- Research: IOS DHCP server pools and excluded address ranges.

Phase 6: Logging and config backups

Goal: the Ubuntu server receives syslog messages from both switches with accurate timestamps, and both running configurations are backed up to it over TFTP.

- Done when: shutting down a port produces a log line on the server, and both config files appear on it.
- Research: enabling remote syslog reception in rsyslog, running a TFTP server on Ubuntu, and adding timestamps to switch log messages.

Phase 7 (optional): Access-layer security

Goal: limit the user port to one learned device with port security, and turn on DHCP snooping on SW2 for VLAN 10, trusting only the uplink toward the DHCP server.

- Done when: plugging in a second device triggers a violation, and the laptop still gets an address.
- Research: port security violation modes, and DHCP option 82 if clients stop getting addresses after snooping is on. Working through that one makes a strong troubleshooting-log entry.

Failure tests

Test	How to run it	What to capture	What you saw
One EtherChannel link fails	Unplug one uplink cable	Lost pings; whether the bundle stays up and which link dropped	
Both uplinks fail and recover	Unplug both, wait, plug both back in	Lost pings; time until traffic resumes	

Test	How to run it	What to capture	What you saw
BPDU Guard trips	Connect a cable from SW2 to a BPDU Guard port on SW1	The log message and the port's state afterward	
Native VLAN mismatch	Change the native VLAN on one side only	Log messages from both switches; trunk status on each end	
Root bridge moves	Make SW2 more preferred than SW1	Which switch is root before and after	
Wrong access VLAN	Move the laptop's port to VLAN 20	What address the laptop gets, if any	
Port security violation	Plug a second device into the secured port	The violation count and log message	

Undo each test before starting the next one. If a port ends up error-disabled, working out how to recover it is part of the exercise.